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Axion Dark Matter

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Outline

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The Strong CP Problem

Symmetries of the Standard Model (SM) allow a topological term

$$\mathcal{L} \supset i\theta \frac{g_s^2}{32\pi^2} G^a_{\mu\nu} \tilde{G}^{\mu\nu a}$$

This term violates CP symmetry, and leads to an electric dipole moment for the neutron (NEDM).

In principle, θ can take any value between 0 and 2π.

Yet, experimental limits on NEDM put it at $\theta \lesssim 10^{-11}$.

This "unnatural" smallness is the Strong-CP Problem.

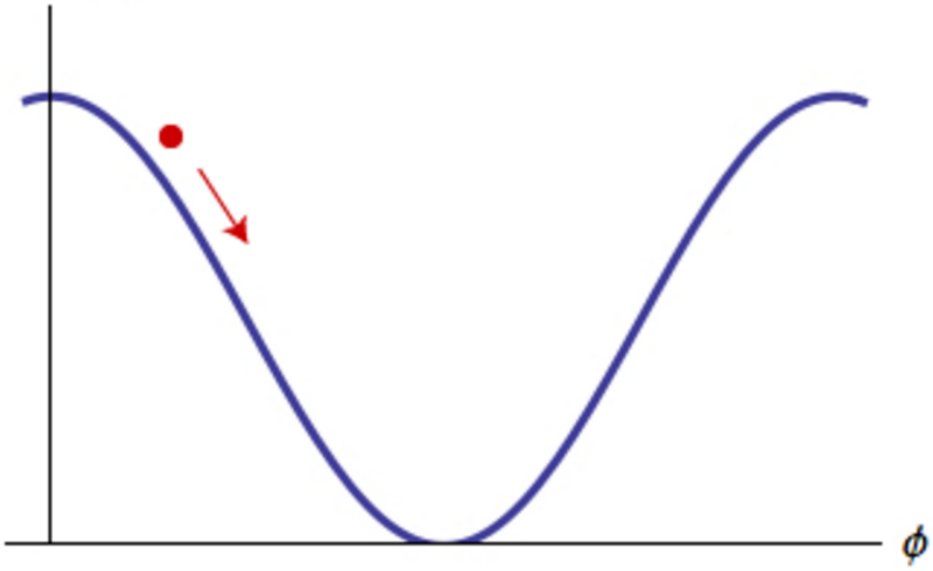
The Peccei-Quinn Mechanism

Peccei and Quinn's idea: restore anomalous $U(1)_A$ symmetry of QCD by introducing a new field.

Essentially they promote parameter θ to a field $a(x)$.

Anomaly effects create a potential proportional to $1 - \cos(\theta)$

Field $a(x)$ "rolls" to the minimum at $\theta = 0$, which conserves CP.



The Peccei-Quinn Mechanism

Wilczek and Weinberg showed that the field introduced in this way is a pseudoscalar, a, and that it also has a mass $m_a \sim 100$ keV.

$$\mathcal{L} \supset \frac{1}{2}\partial_\mu a \partial^\mu a + \frac{g_s^2}{32\pi^2}\left(\frac{a}{F_a} + \theta'\right) G^a_{\mu\nu}\tilde{G}^{\mu\nu a}$$

They showed that in the simple models, the axion could be part of the Higgs sector in the Standard Model

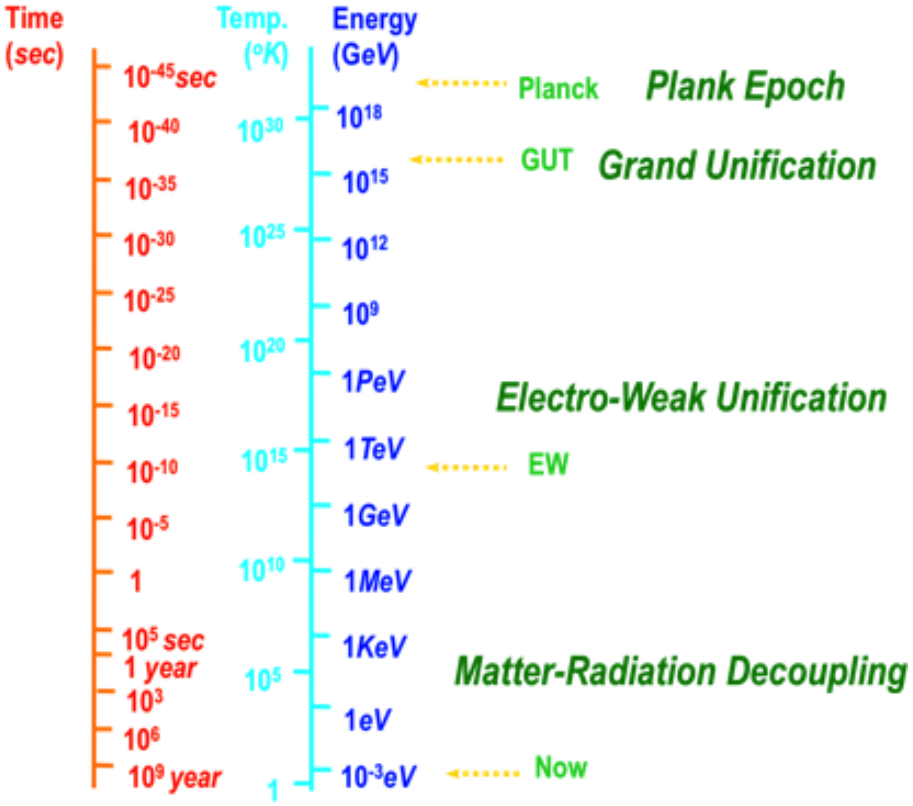
This sets "scale" of Axion interactions at the electroweak scale, $v = 246$ GeV, and makes robust predictions that are testable in e.g., reactor experiments.

Invisible Axions

Simple "Higgs"-like axion models were quickly ruled out, so theorists had to separate F_a from the electroweak scale.

If F_a is large enough, then the axion can evade experimental and astrophysical constraints.

The natural assumption was that axions would be part of a Grand Unified Theory, so $F_a \sim 10^{15}$ GeV.



home.physics.ucla.edu/~arisaka/home/Particle/

Invisible Axion Models

Two main classes of models:

- **Hadronic (KSVZ) models:** Introduce new pair of quarks Q, $\bar{Q}$

$$\mathcal{L} \supset k\phi Q\bar{Q}$$

- **DFSZ models:** Couple ϕ to two standard model Higgs doublets H_1 and H_2

$$\begin{aligned}\mathcal{L} \quad \supset \quad & |\phi|^2\left(c_1|H_1|^2 + c_2|H_2|^2\right) + c_3|H_1|^2|H_2|^2 + c_4|H_1H_2|^2 \\ & + \left(\mu\phi H_1 H_2 + \text{h.c.}\right),\end{aligned}$$

All models universally characterized by PQ scale F_a.

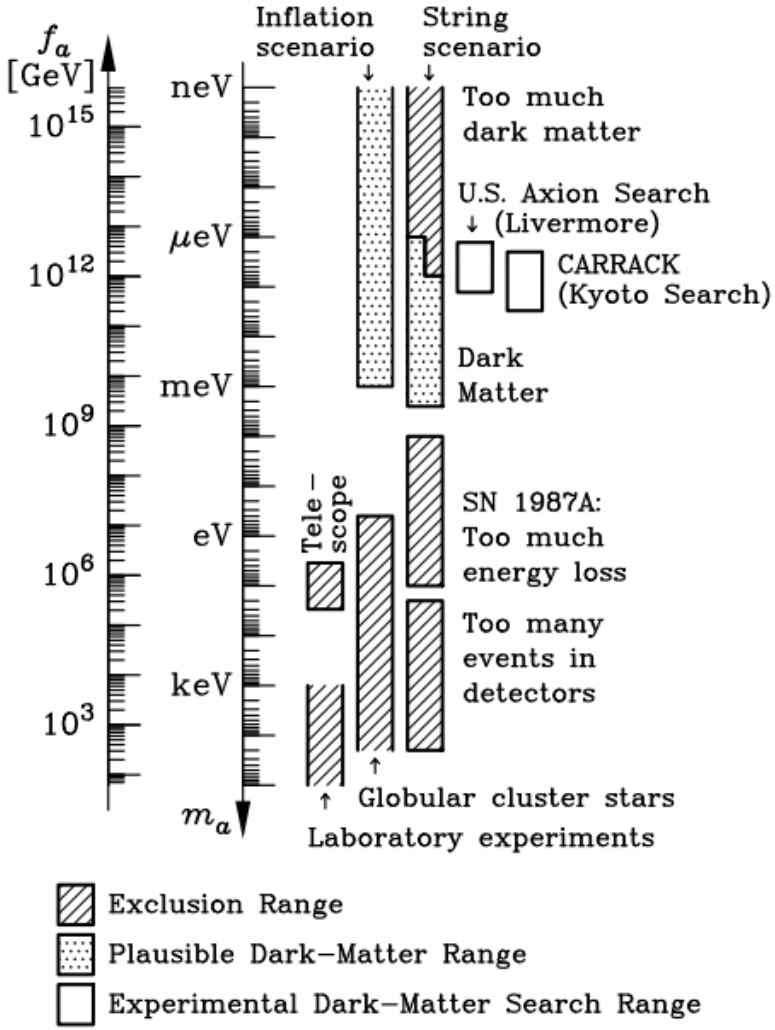
Astrophysical Limits

Because all axion models can be characterized by F_a, we can look for "universal" constraints from a variety of contexts.

Early stringent constraints came from evolution of Red Giant (He) star evolution: axions not affecting growth requires $F_a \gtrsim 10^9$ GeV.

To ensure that they don't modify burst duration of SN1987A sets $F_a \geq 4 \times 10^8$ GeV.

These constraints seem consistent with GUT ideas of $F_a \sim 10^{15}$ GeV.



arXiv:hep-ph/9903472

Axion Energy Density

A trio of papers in 1983 show that there is a cosmological upper bound due to the energy density.

The lifetime of $a \to \gamma\gamma$ decays is $\sim 10^{24}$ yr, so energy cannot dissipate through particle decays.

After spontaneous breaking of Peccei-Quinn symmetry,

$$\frac{d^2\langle a\rangle}{dt^2} + 3H(t)\frac{d\langle a\rangle}{dt} + m_a^2(t)\langle a\rangle = 0.$$

Using the solution $\langle a\rangle = A(t)\cos(m_a t)$, we can compute the energy density:

$$\Omega_a = \left(\frac{10^7 F_a}{M_{\rm Pl}}\right)\left(\frac{200\ \text{GeV}}{T_i}\right)\left(\frac{A_i}{F_a}\right)^2 h^{-2}$$

Requiring Ω_a less than critical density constrains $F_a \lesssim 4 \times 10^{12}$ GeV.

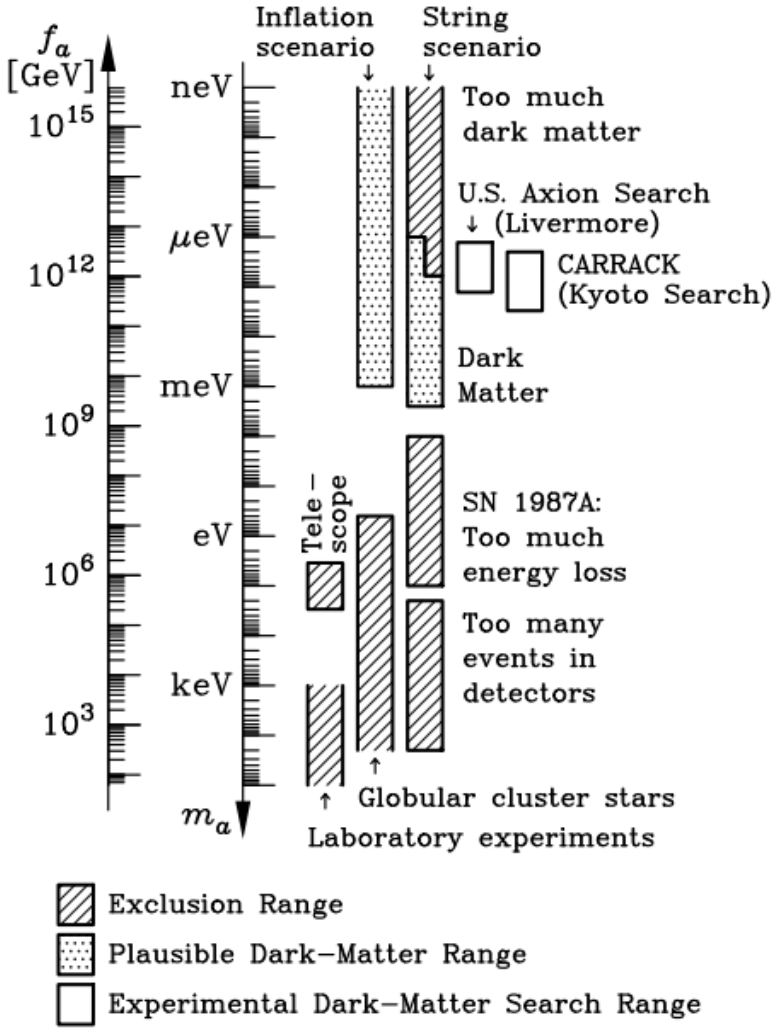
Axions as Dark Matter

The original authors interpreted bound $F_a \lesssim 10^{12}$ GeV as problem with fine-tuning for GUTs.

But they also noted that if $F_a \sim 10^{10} - 10^{11}$ GeV, then Ω_a is of the order the dark matter density!

Axion "gas" is non-dissipative and pressureless: behaves like dust, and fluctuations can grow like normal matter.

$\Longrightarrow$ The axion is a natural warm dark matter candidate.



arXiv:hep-ph/9903472

The Inverse-Primakoff Effect

Axions couple to electromagnetic fields, with coupling $\propto F_a^{-1}$

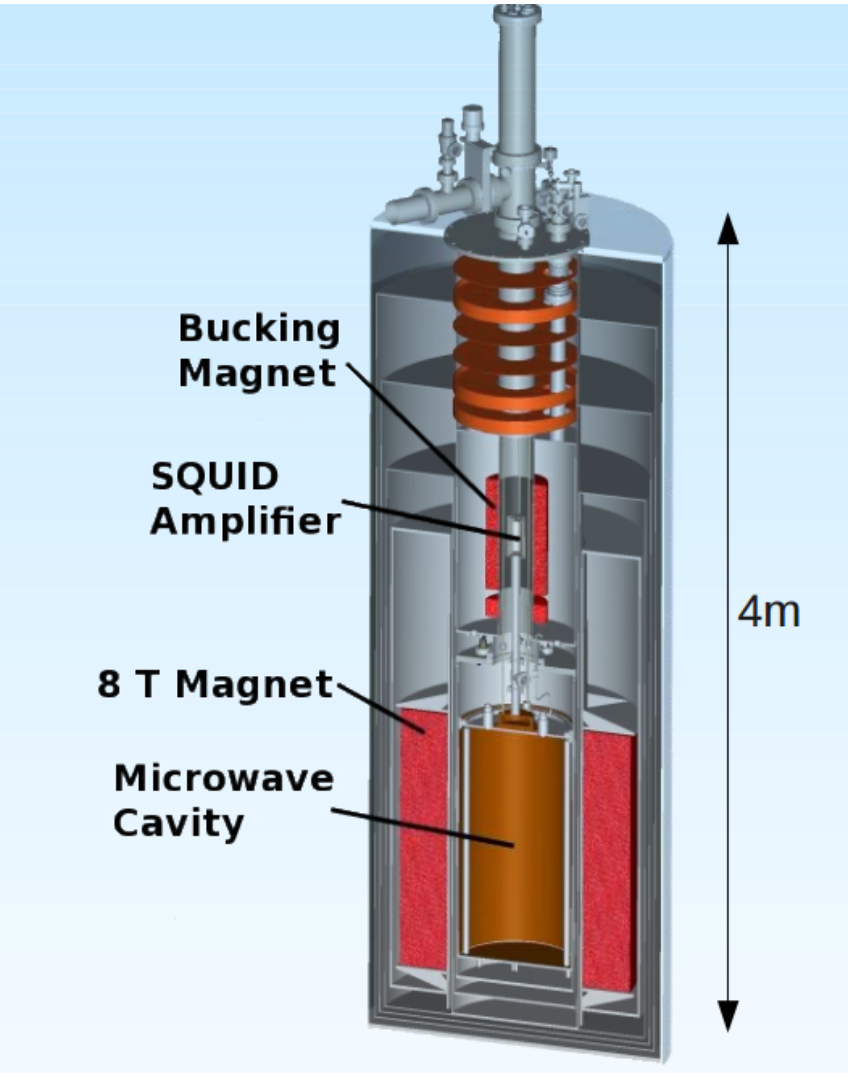
Classical equations of motion:

$$\vec{\nabla} \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \frac{e^2 g_\gamma}{4\pi^2 F_a} \left(\vec{E} \times \vec{\nabla} a - \vec{B} \frac{\partial a}{\partial t} \right),$$

$\implies$ axions in a strong, inhomogeneous magnetic field will convert to photons.

Idea: Search for dark matter halo axions interacting with magnetic field by magnifying effect in resonant cavity.

The ADMX Experiment



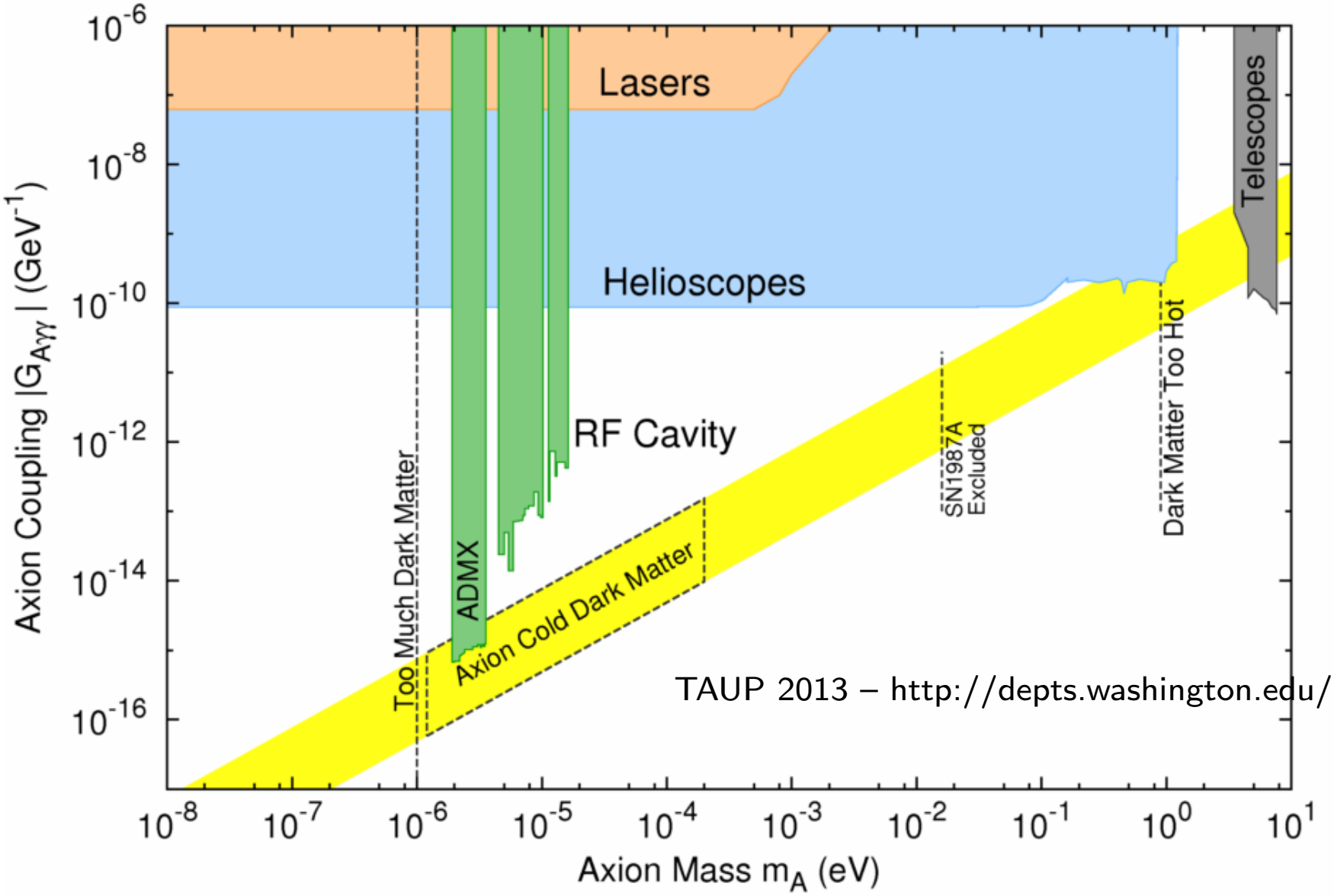
courtesy: depts.washington.edu/admx/

ADMX uses microwave cavity as source of virtual photons that can interact with DM halo axions.

SQUID amplifier gives them the sensitivity to probe most of the interesting dark matter parameter space.

Have started taking science data.. waiting for results.

Allowed Parameter Space



TAUP 2013 – http://depts.washington.edu/admx

Conclusions and Outlook

- The Peccei-Quinn mechanism, leading to axions, is a powerful tool for solving the strong CP problem.
- Axions are an interesting candidate for dark matter in a variety of theories.
- Axions can be detected in a variety of model dependent ways, including haloscopes, helioscopes, and cold atom experiments.

Outlook:

- The presence of axions must be considered in particle physics model building.
- Upcoming experiments will probe the most interesting parameter space of axion models.